CO2xN: Porewater Sulfide

Month Year Samples

2026-02-17

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed

sample_month = "May"
sample_year = "2025"
Run_Date = as.Date("05/27/2025", format = "%m/%d/%Y")
Run_by = "Melanie Giessner"#Instrument user
Script_run_by ="Brenden Blakley"#Code user

Run_notes=""#any notes from run

#Import Plate Data

plates<- c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7", "Plate8", "Plate9", "Plate10",

Raw_plates <- c("Raw Data/20250714_LTREB_H2S_Plate1.csv",
                "Raw Data/20250714_LTREB_H2S_Plate2.csv",
                "Raw Data/20250714_LTREB_H2S_Plate3.csv",
                "Raw Data/20250714_LTREB_H2S_Plate4.csv",
                "Raw Data/20250714_LTREB_H2S_Plate5.csv",
                "Raw Data/20250714_LTREB_H2S_Plate6.csv",
                "Raw Data/20250715_LTREB_H2S_Plate7.csv",
                "Raw Data/20250715_LTREB_H2S_Plate8.csv",
                "Raw Data/20250715_LTREB_H2S_Plate9.csv",
                "Raw Data/20250715_LTREB_H2S_Plate10.csv",
                "Raw Data/20250716_LTREB_H2S_reruns.csv")

#Import Run Date for Each Plate for Std curve qaqc
Run_dates <- tibble(
    Plate = c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7","Plate8","Plate9","Plate10",
    Run_Date = c("07/14/2025","07/14/2025","07/14/2025","07/14/2025","07/14/2025","07/14/2025","07/15/2025
  )

  Run_dates <- Run_dates %>% mutate(Run_Date = as.Date(Run_Date, format = "%m/%d/%Y"))


# Define the file path for QAQC log file - NO Need to change just check year
```

```r
log_path <- "Processed Data/LTREB_2M_H2S_QAQC_2025.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202507.csv"
rerun_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202507_reruns.csv"
```

##Read in metadata and create similar sample IDs for matching to samples

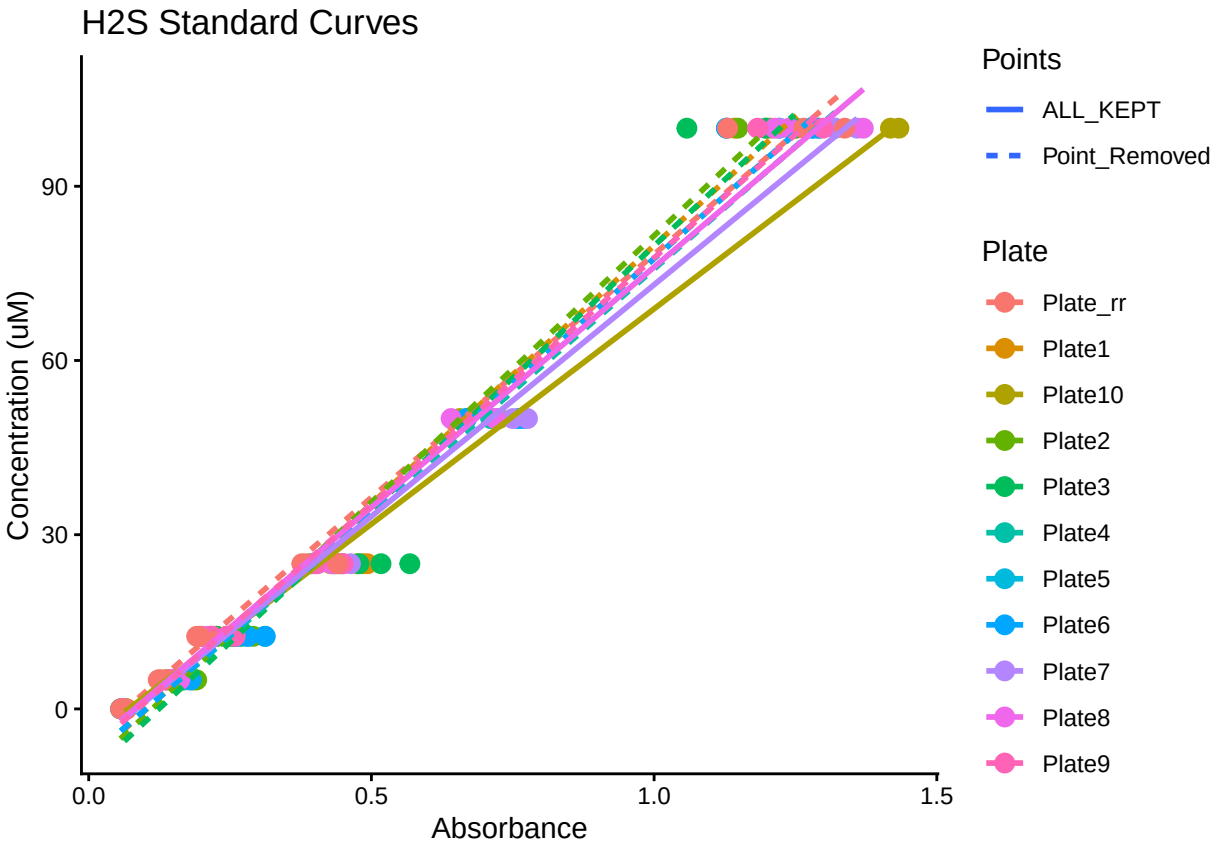
##Read in data

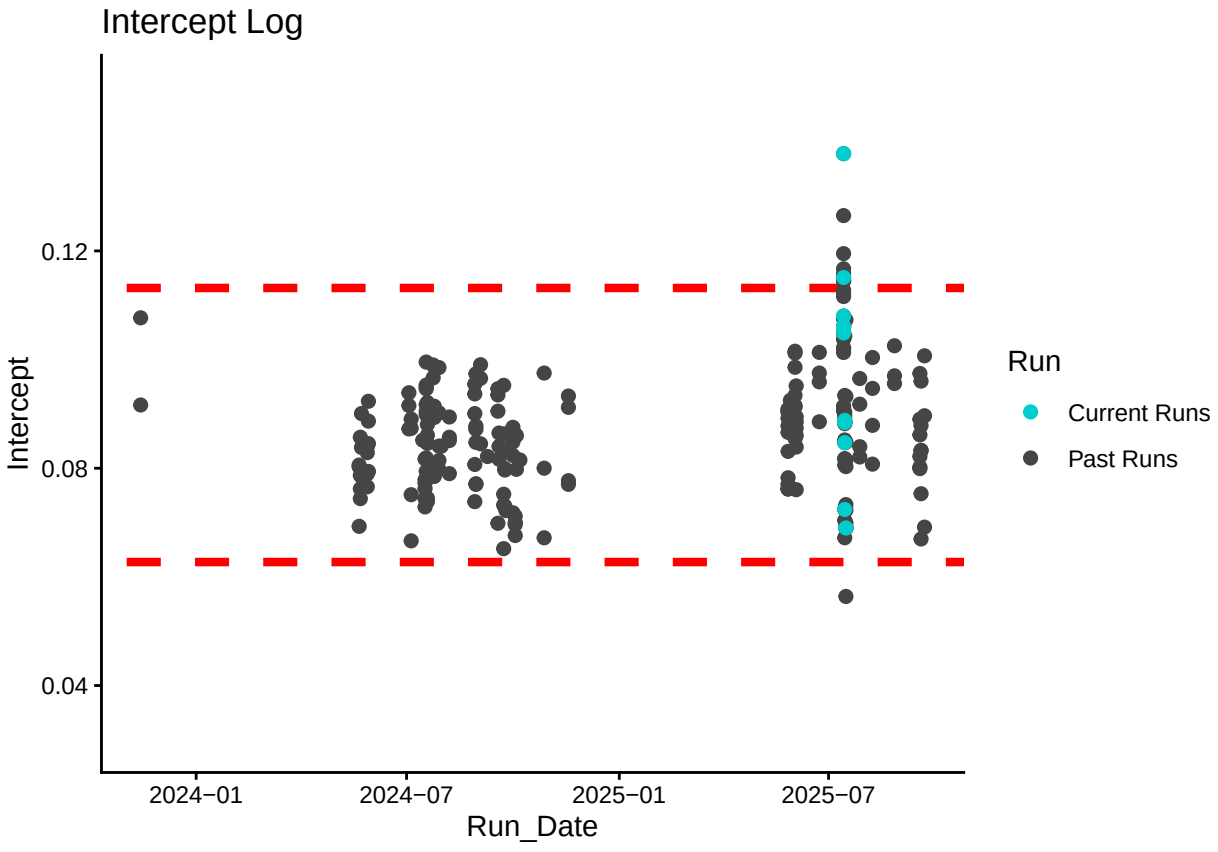
##Remove High CV Stds

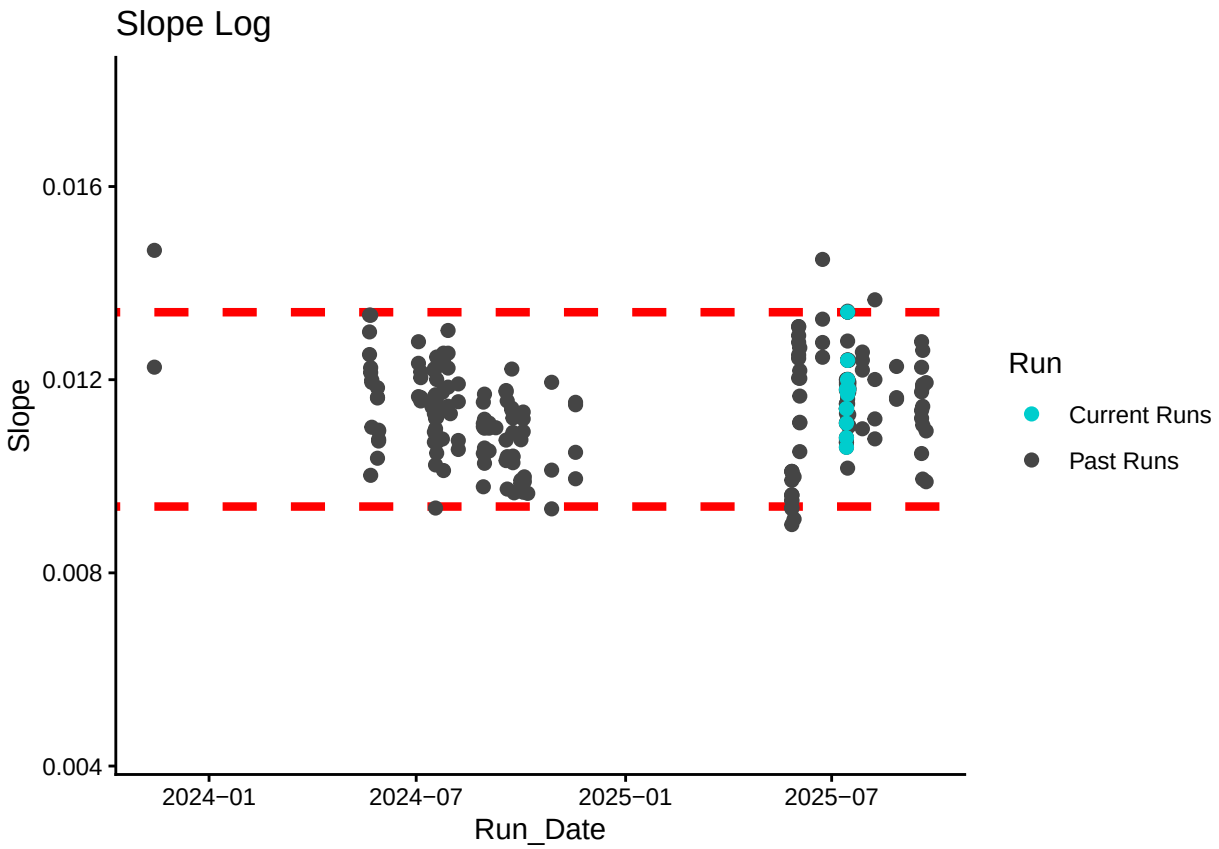
##Check R2 & Remove Standards if Necessary

```
## Joining with `by = join_by(Plate, IDs)`
```

Plot Standard Curves







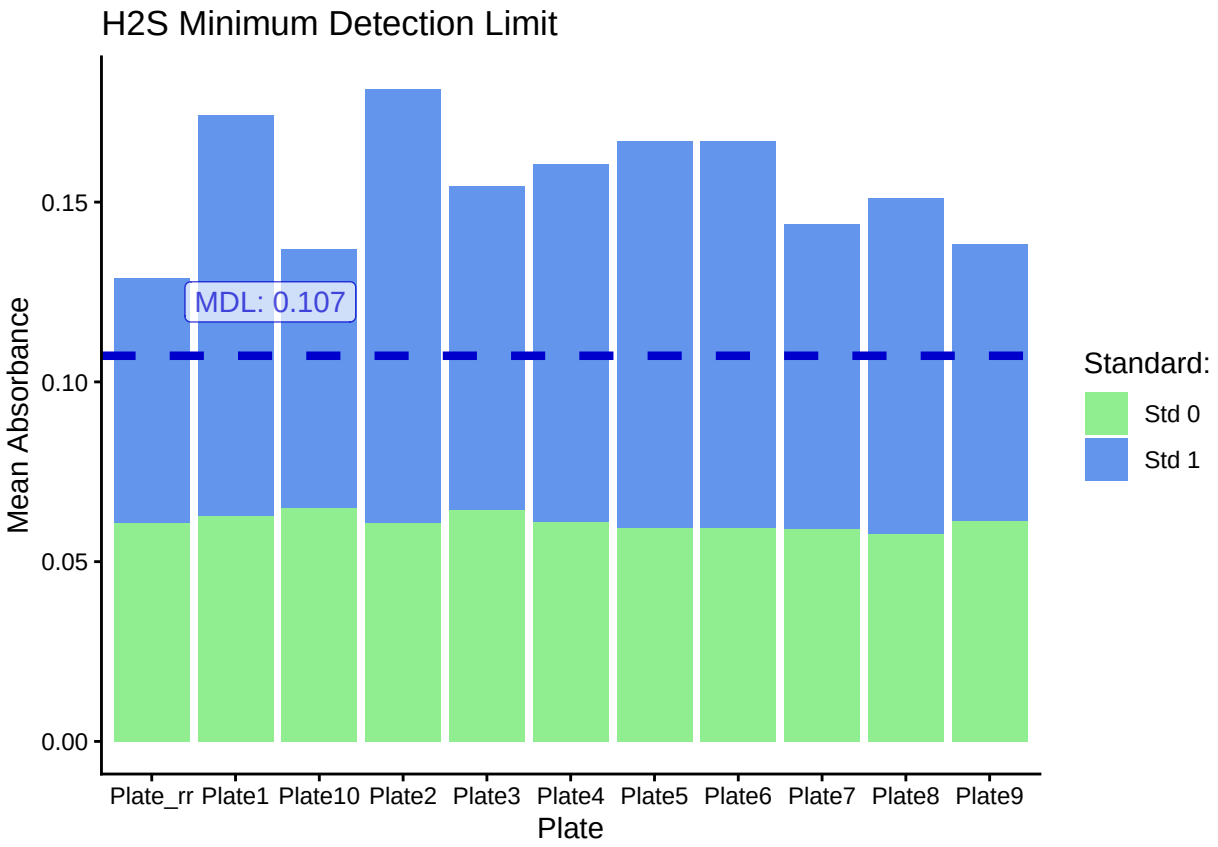
Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Bad	CV_Good
Plate10	R2_Good	CV_Good
Plate2	R2_Good	CV_Good
Plate3	R2_Bad	CV_Good
Plate4	R2_Good	CV_Good
Plate5	R2_Good	CV_Good
Plate6	R2_Good	CV_Good
Plate7	R2_Good	CV_Good
Plate8	R2_Good	CV_Good
Plate9	R2_Good	CV_Good
Plate_rr	R2_Good	CV_Good

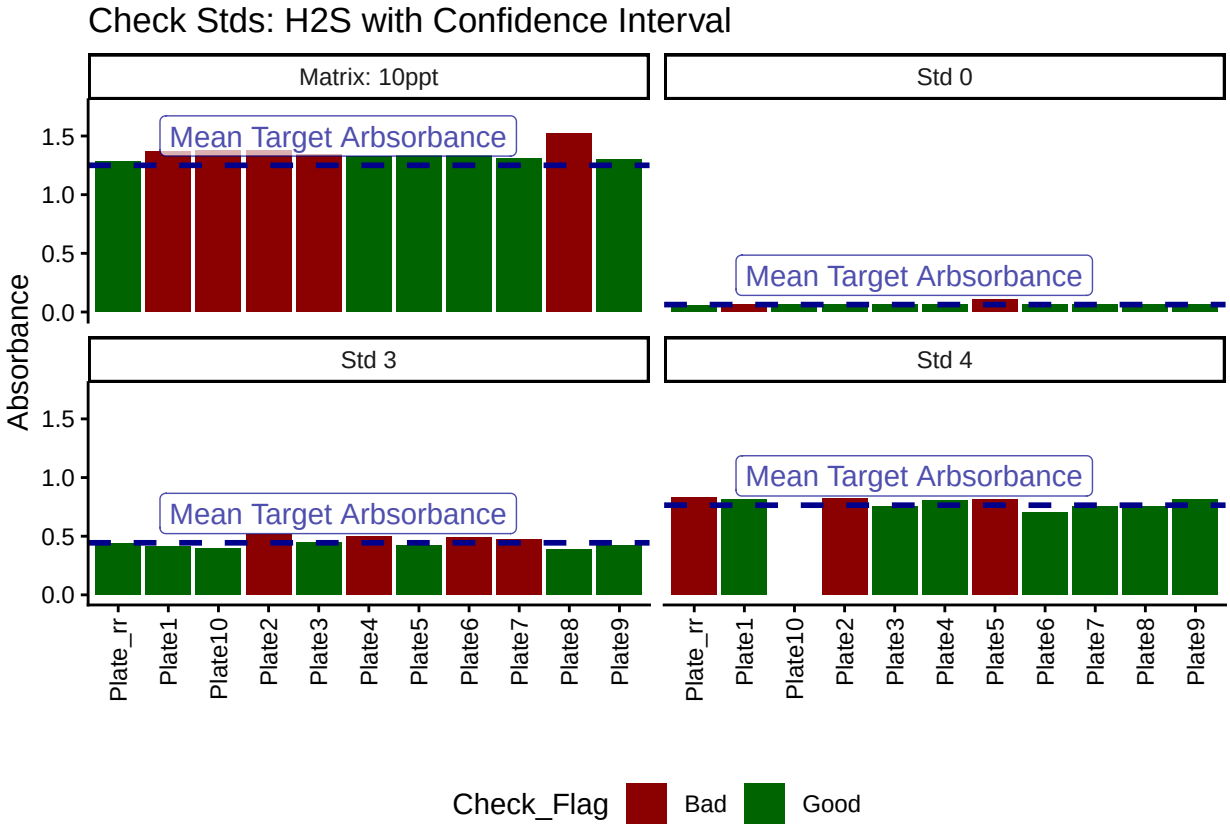
Plate	Curve_RMVD
Plate1	KEPT
Plate2	REMOVED
Plate3	REMOVED
Plate4	KEPT
Plate5	KEPT
Plate6	KEPT
Plate7	KEPT
Plate8	KEPT
Plate9	KEPT
Plate10	REMOVED
Plate_rr	KEPT

Plate	Curve_RMVD

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

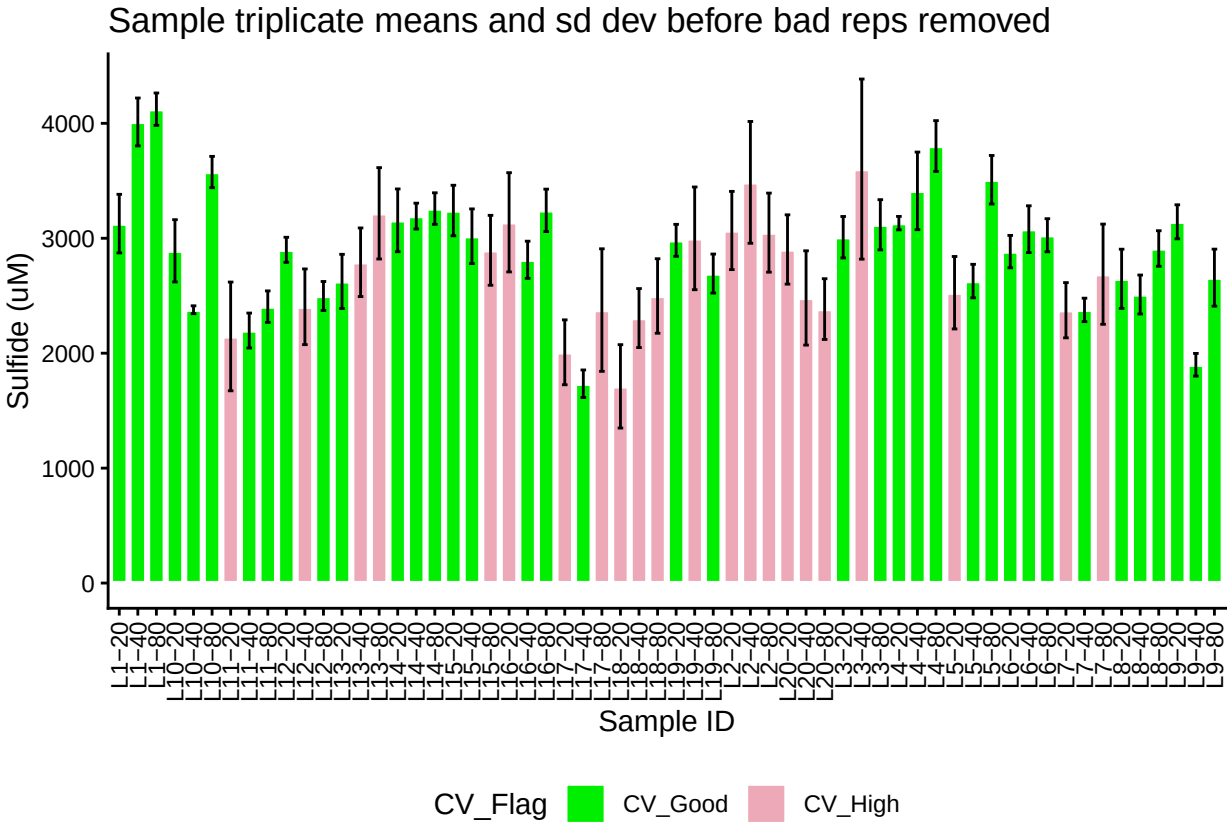
Plate	CHKs_Flag
Plate1	Std0Chk_Bad
Plate1	Std5Chk_Bad
Plate2	Std3Chk_Bad
Plate2	Std5Chk_Bad
Plate2	Std4Chk_Bad
Plate3	Std5Chk_Bad
Plate4	Std3Chk_Bad
Plate5	Std0Chk_Bad
Plate5	Std4Chk_Bad
Plate6	Std3Chk_Bad
Plate7	Std3Chk_Bad
Plate8	Std5Chk_Bad
Plate10	Std5Chk_Bad
Plate_rr	Std4Chk_Bad

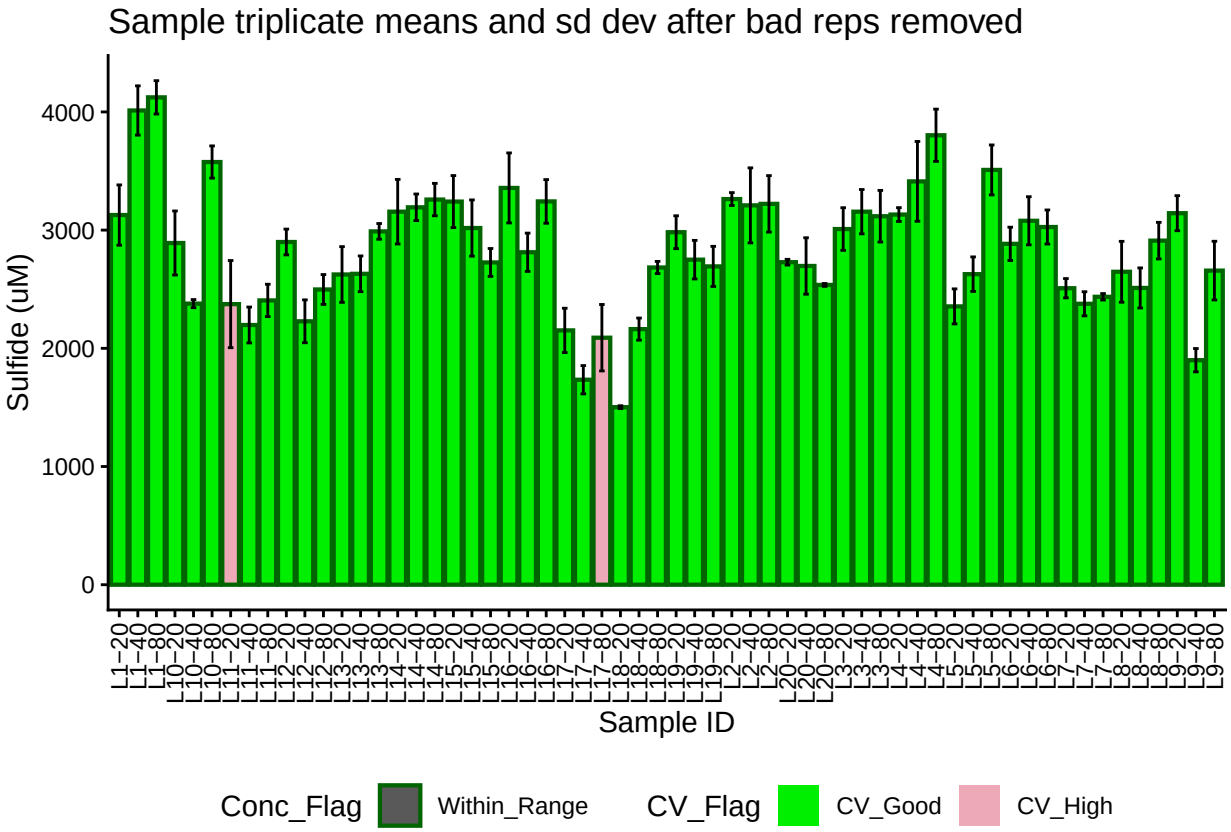
Plate	CHK_CV_Flags
Plate5	Std0Chk_CV_High

##Calculate Sulfide Concentrations & Add Concentration Flags

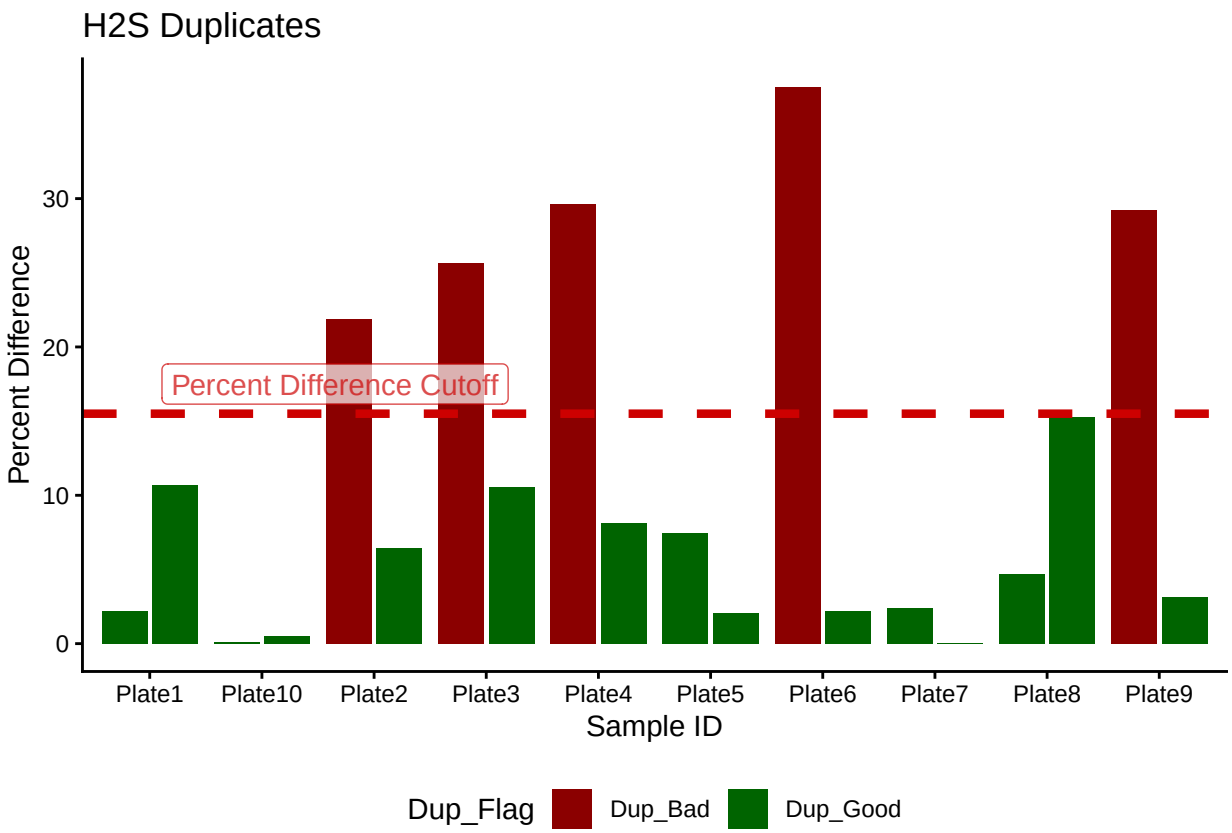
##Remove High CV Samples and Average

Plot Samples





Duplicate QAQC

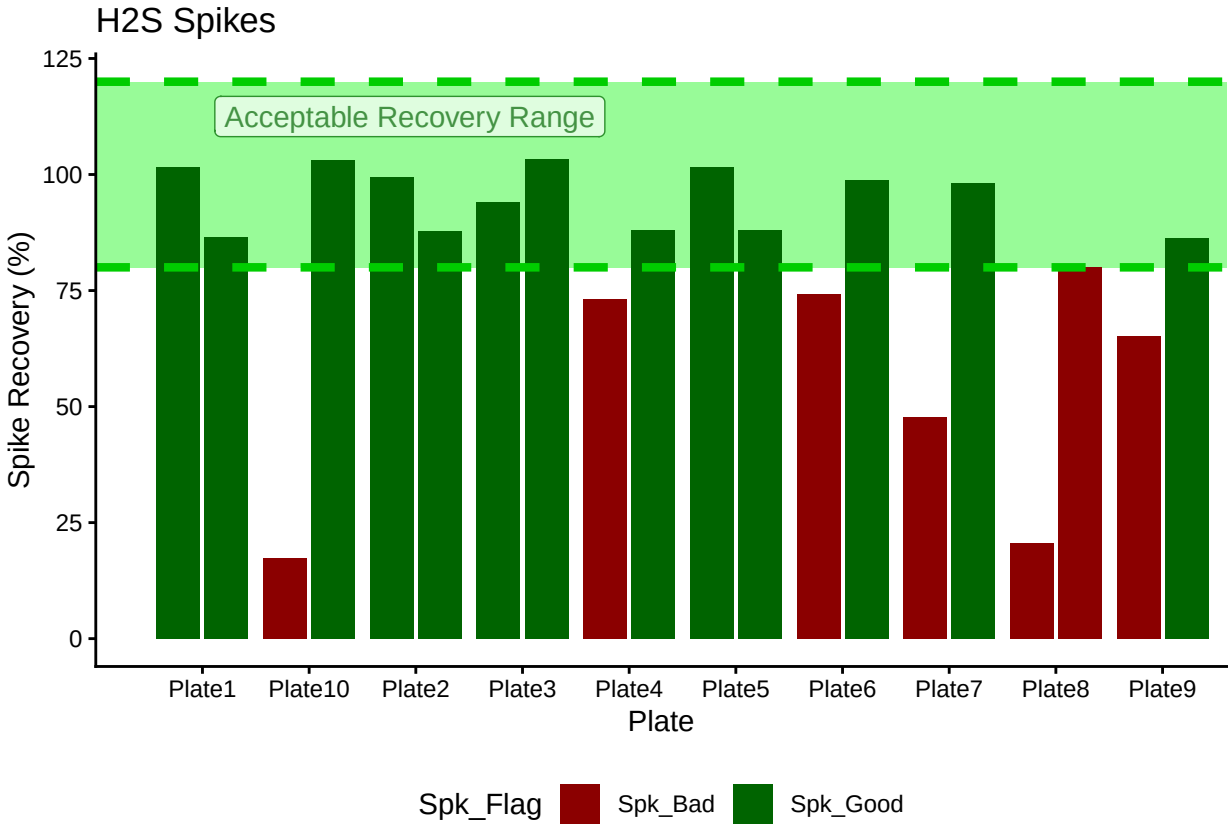


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Bad
Plate3	Dup_Bad
Plate4	Dup_Bad
Plate5	Dup_Good
Plate6	Dup_Bad
Plate7	Dup_Good
Plate8	Dup_Good
Plate9	Dup_Bad
Plate10	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Good
Plate2	Spk_Good
Plate3	Spk_Good
Plate4	Spk_Bad
Plate5	Spk_Good
Plate6	Spk_Bad
Plate7	Spk_Bad
Plate8	Spk_Bad
Plate9	Spk_Bad
Plate10	Spk_Bad

```
## [1] "Spikes Pass"
```

Merge samples with Metadata & check all samples are present

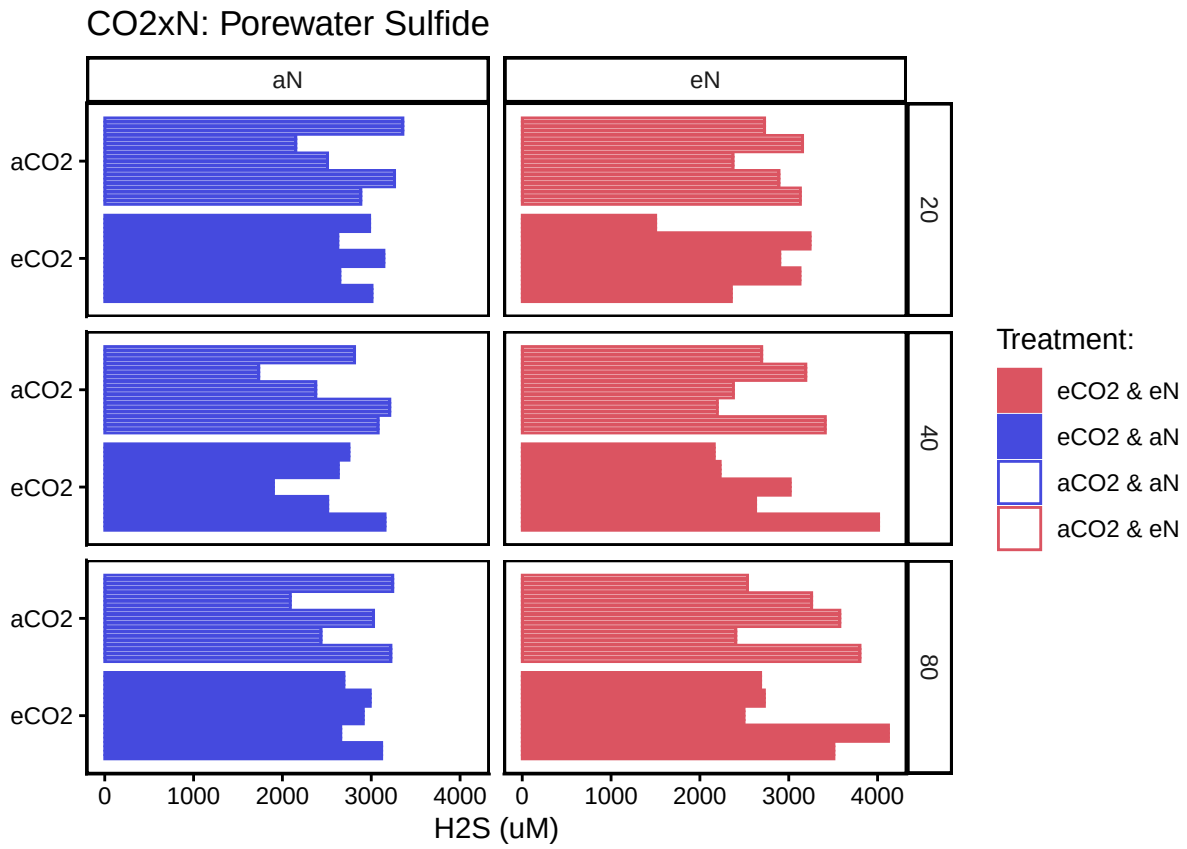
```
## All sample IDs are present in data
```

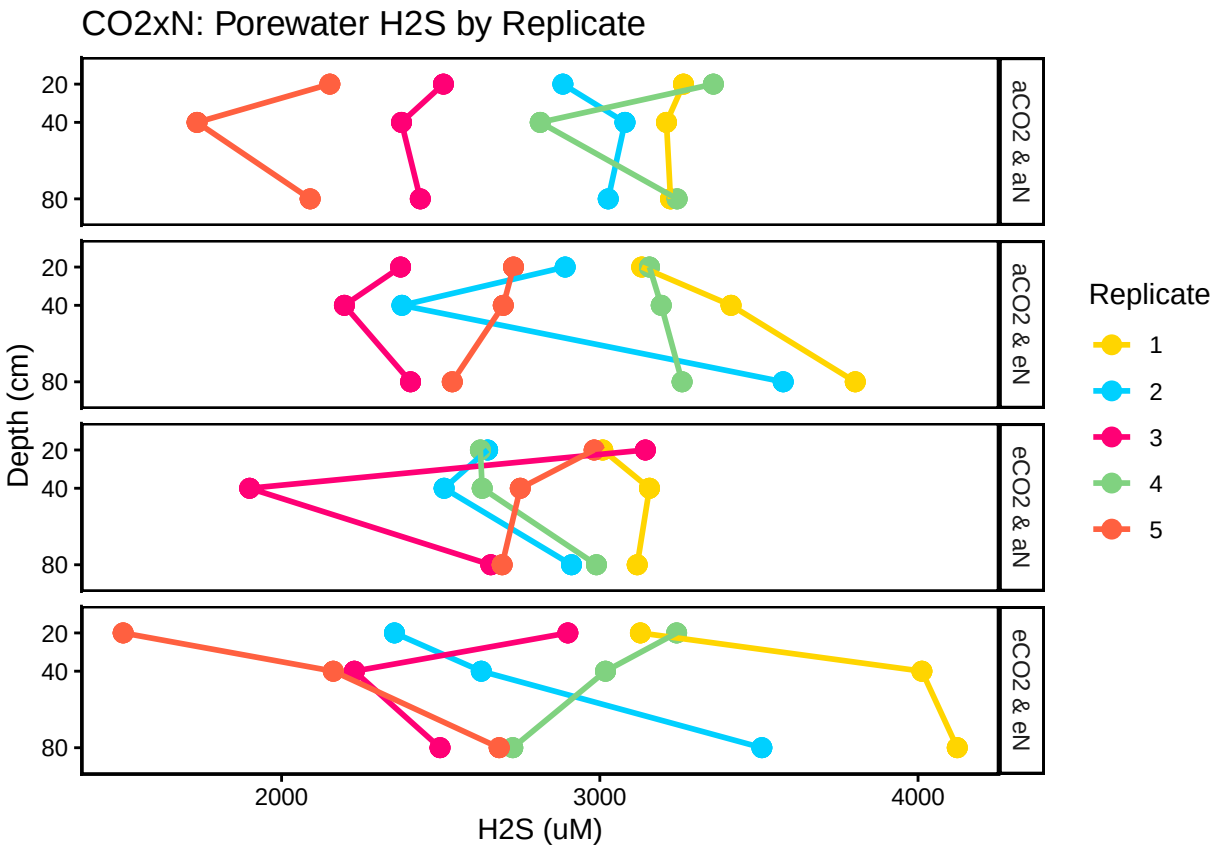
##Add Flags

```
## Warning in left_join(merged_data, Flags, by = join_by(Plate)): Detected an unexpected many-to-many rela
## i Row 1 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 1 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```

##Organize Data

Visualize Data





##Export Data

END